

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Saturday, 22 August 2026

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Integrals

1.1 Medium Integral

Rational Function with Golden Ratio

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \frac{x^2}{x^2 + (x^2 - \phi)^2} dx$$

Solution: Let I be the given integral. We can begin by expanding the polynomial expression in the denominator:

$$x^2 + (x^2 - \phi)^2 = x^2 + x^4 - 2\phi x^2 + \phi^2 = x^4 + (1 - 2\phi)x^2 + \phi^2$$

This transforms our integral into a standard rational form:

$$I = \int_{-\infty}^{\infty} \frac{x^2}{x^4 + (1 - 2\phi)x^2 + \phi^2} dx$$

For any integral of the general form $\int_{-\infty}^{\infty} \frac{x^2}{x^4 + bx^2 + c^2} dx$ where $c > 0$ and $b > -2c$, the value of the integral is given by the established formula:

$$\frac{\pi}{\sqrt{2c + b}}$$

In our case, we can map the coefficients directly: $c = \phi$ (since the golden ratio $\phi > 0$) and $b = 1 - 2\phi$. We must first verify the convergence condition $b > -2c$:

$$1 - 2\phi > -2\phi \implies 1 > 0$$

The condition holds beautifully. Substituting our coefficients into the formula gives:

$$I = \frac{\pi}{\sqrt{2(\phi) + (1 - 2\phi)}} = \frac{\pi}{\sqrt{2\phi + 1 - 2\phi}} = \frac{\pi}{\sqrt{1}}$$

Strikingly, the golden ratio cancels out entirely, yielding:

$$I = \pi$$

1.2 Hard Integral

Oscillatory Exponential Integral

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \frac{e^{\cos(x)} \sin(x - \sin(x))}{x^3} dx$$

Solution: Let the integral be I . First, we recognize that the integrand is an even function, and there is no singularity at $x = 0$ due to the Taylor series expansion of $\sin(x - \sin(x)) \approx \frac{x^3}{6}$.

We can elegantly represent the numerator as the imaginary part of a complex exponential. Consider the function $e^{ix+e^{-ix}}$:

$$e^{ix+e^{-ix}} = e^{ix} e^{\cos(x)-i\sin(x)} = e^{\cos(x)} e^{i(x-\sin(x))}$$

By taking the imaginary part, we recover exactly our numerator:

$$\Im(e^{ix+e^{-ix}}) = e^{\cos(x)} \sin(x - \sin(x))$$

Thus, we rewrite the integral as:

$$I = \Im \int_{-\infty}^{\infty} \frac{e^{ix+e^{-ix}}}{x^3} dx$$

We expand $e^{e^{-ix}}$ using its Maclaurin series $\sum_{k=0}^{\infty} \frac{e^{-ikx}}{k!}$:

$$I = \Im \sum_{k=0}^{\infty} \frac{1}{k!} \int_{-\infty}^{\infty} \frac{e^{i(1-k)x}}{x^3} dx$$

Taking the imaginary part gives us the sine function:

$$I = \sum_{k=0}^{\infty} \frac{1}{k!} \int_{-\infty}^{\infty} \frac{\sin((1-k)x)}{x^3} dx$$

Using the known principal value result $\int_{-\infty}^{\infty} \frac{\sin(ax)-ax}{x^3} dx = -\frac{\pi}{2}a^2 \operatorname{sgn}(a)$ (where the auxiliary $-ax$ terms sum to zero across the series), we substitute $a = 1 - k$:

$$I = \sum_{k=0}^{\infty} \frac{1}{k!} \left(-\frac{\pi}{2} (1-k)^2 \operatorname{sgn}(1-k) \right)$$

We evaluate this sum piecewise based on the value of k : For $k = 0$, $a = 1$, the term is $-\frac{\pi}{2}$. For $k = 1$, $a = 0$, the term is 0. For $k \geq 2$, $1 - k < 0$, so $\operatorname{sgn}(1 - k) = -1$. The terms become $\frac{\pi}{2} \frac{(k-1)^2}{k!}$. Summing from $k = 2$ to ∞ :

$$\sum_{k=2}^{\infty} \frac{(k-1)^2}{k!} = \sum_{k=2}^{\infty} \frac{k(k-1) - k + 1}{k!} = \sum_{k=2}^{\infty} \frac{1}{(k-2)!} - \sum_{k=2}^{\infty} \frac{1}{(k-1)!} + \sum_{k=2}^{\infty} \frac{1}{k!}$$

These are standard shifted exponential series summing to e , $(e - 1)$, and $(e - 2)$ respectively:

$$e - (e - 1) + (e - 2) = e - 1$$

Putting it all together:

$$I = -\frac{\pi}{2} + \frac{\pi}{2}(e - 1) = \frac{\pi}{2}(e - 2)$$

Differentials

2.1 Medium Differential

Wronskian First-Order Equation

Consider the following differential equation:

$$\frac{dy}{dx} = W_x(\alpha, \beta, \gamma)$$

where $W_x(\alpha, \beta, \gamma)$ is the Wronskian determinant with respect to x (treating y as a constant) and

$$\alpha = \frac{1}{y}, \quad \beta = \frac{x}{y}, \quad \gamma = \frac{x^2 e^{\pi y}}{2}.$$

Using the given initial condition $y \left(\frac{(\pi \ln(\pi))^2 + 2\pi \ln(\pi) + 2}{-\pi^3 + \pi} \right) = \ln(\pi)$, calculate $-x\pi^3 e$ when $y = \frac{1}{\pi}$.

Solution: We compute the Wronskian $W_x(\alpha, \beta, \gamma)$ with respect to x . The functions and their derivatives are: $\alpha = y^{-1}$, $\alpha' = 0$, $\alpha'' = 0$, $\beta = xy^{-1}$, $\beta' = y^{-1}$, $\beta'' = 0$, $\gamma = \frac{1}{2}x^2 e^{\pi y}$, $\gamma' = xe^{\pi y}$, $\gamma'' = e^{\pi y}$. Since $\alpha' = \alpha'' = 0$, the Wronskian simplifies to:

$$W_x = \alpha(\beta'\gamma'' - \beta''\gamma') = \frac{1}{y} \left(\frac{1}{y} e^{\pi y} - 0 \right) = \frac{e^{\pi y}}{y^2}$$

The differential equation is thus $\frac{dy}{dx} = \frac{e^{\pi y}}{y^2}$. Separating variables:

$$y^2 e^{-\pi y} dy = dx$$

Integrating both sides (using integration by parts twice on the left):

$$\int y^2 e^{-\pi y} dy = -e^{-\pi y} \left(\frac{y^2}{\pi} + \frac{2y}{\pi^2} + \frac{2}{\pi^3} \right)$$

So, $x = -e^{-\pi y} \left(\frac{\pi^2 y^2 + 2\pi y + 2}{\pi^3} \right) + C$. Using the complex initial condition where $y = \ln(\pi)$, we see that plugging $y = \ln(\pi)$ into our integrated expression identically matches the given x -coordinate, demanding that $C = 0$. We want to find $-x\pi^3 e$ when $y = \frac{1}{\pi}$. First, substitute $y = \frac{1}{\pi}$:

$$x = -e^{-1} \left(\frac{\pi^2 (1/\pi^2) + 2\pi (1/\pi) + 2}{\pi^3} \right) = -\frac{1}{e} \left(\frac{1 + 2 + 2}{\pi^3} \right) = -\frac{5}{e\pi^3}$$

Finally, evaluating the target expression:

$$-x\pi^3 e = - \left(-\frac{5}{e\pi^3} \right) \pi^3 e = 5$$

2.2 Hard Differential

Trigonometric Riccati Equation

Solve for $0 \leq x \leq \frac{\pi}{4}$:

$$\frac{dy}{dx} = e^x + \tan(x)y - e^{-x} \tan(x)y^2$$

Given $y(0) = 3$, find the exact value of $y\left(\frac{\pi}{4}\right)$.

Solution: This is a Riccati equation. We can find a particular solution by inspection. Testing $y_1(x) = e^x$:

$$y_1' = e^x \implies e^x = e^x + \tan(x)e^x - e^{-x} \tan(x)e^{2x} = e^x$$

Since it satisfies the equation, we use the standard Riccati substitution $y = e^x + \frac{1}{u}$, which gives $y' = e^x - \frac{u'}{u^2}$. Substituting this into the ODE simplifies the system to a linear differential equation in u :

$$e^x - \frac{u'}{u^2} = e^x + \tan(x) \left(e^x + \frac{1}{u} \right) - e^{-x} \tan(x) \left(e^{2x} + \frac{2e^x}{u} + \frac{1}{u^2} \right)$$

Canceling terms yields:

$$-\frac{u'}{u^2} = -\frac{\tan(x)}{u} - \frac{e^{-x} \tan(x)}{u^2} \implies u' - \tan(x)u = e^{-x} \tan(x)$$

The integrating factor is $\mu(x) = e^{\int -\tan(x) dx} = \cos(x)$. Multiplying through:

$$(u \cos(x))' = e^{-x} \sin(x)$$

Integrating the right-hand side via parts:

$$u \cos(x) = \int e^{-x} \sin(x) dx = -\frac{e^{-x}}{2} (\sin(x) + \cos(x)) + C$$

$$u(x) = C \sec(x) - \frac{e^{-x}}{2} (\tan(x) + 1)$$

Using the initial condition $y(0) = 3$, we have $3 = e^0 + \frac{1}{u(0)} \implies u(0) = \frac{1}{2}$.

$$u(0) = C(1) - \frac{1}{2}(0 + 1) = C - \frac{1}{2} = \frac{1}{2} \implies C = 1$$

So, $u(x) = \sec(x) - \frac{e^{-x}}{2} (\tan(x) + 1)$. To find $y\left(\frac{\pi}{4}\right)$, we first calculate $u\left(\frac{\pi}{4}\right)$:

$$u\left(\frac{\pi}{4}\right) = \sqrt{2} - \frac{e^{-\pi/4}}{2} (1 + 1) = \sqrt{2} - e^{-\pi/4}$$

Finally, reverting the substitution gives:

$$y\left(\frac{\pi}{4}\right) = e^{\pi/4} + \frac{1}{\sqrt{2} - e^{-\pi/4}}$$

Derivatives

3.1 Medium Derivative

Second Derivative of Integral Function

Let $f(x) = \int_0^x x \sin(t^2) dt$. Find $f''(1)$.

Solution: We can factor x out of the integral since the integration variable is strictly t :

$$f(x) = x \int_0^x \sin(t^2) dt$$

Applying the product rule and the Fundamental Theorem of Calculus to find the first derivative:

$$f'(x) = 1 \cdot \int_0^x \sin(t^2) dt + x \cdot \frac{d}{dx} \left(\int_0^x \sin(t^2) dt \right)$$

$$f'(x) = \int_0^x \sin(t^2) dt + x \sin(x^2)$$

Differentiating a second time gives:

$$f''(x) = \sin(x^2) + \left(1 \cdot \sin(x^2) + x \cdot 2x \cos(x^2) \right)$$

$$f''(x) = 2 \sin(x^2) + 2x^2 \cos(x^2)$$

Finally, evaluating this expression at $x = 1$:

$$f''(1) = 2 \sin(1) + 2 \cos(1)$$

3.2 Hard Derivative

Trigonometric Logarithmic Derivative

Let $f(x) = \ln(\sin(x)) \left(\frac{1 - \cos(x) + \sin(x)}{1 + \cos(x) + \sin(x)} \right)^2$. Find $f' \left(\frac{\pi}{3} \right)$.

Solution: We begin by simplifying the rational trigonometric fraction using half-angle substitutions. Using the identities $1 - \cos(x) = 2 \sin^2 \left(\frac{x}{2} \right)$ and $1 + \cos(x) = 2 \cos^2 \left(\frac{x}{2} \right)$, along with $\sin(x) = 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)$, we rewrite the term:

$$\frac{2 \sin^2 \left(\frac{x}{2} \right) + 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{2 \cos^2 \left(\frac{x}{2} \right) + 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}$$

Factoring the numerator and denominator:

$$\frac{2 \sin \left(\frac{x}{2} \right) \left[\sin \left(\frac{x}{2} \right) + \cos \left(\frac{x}{2} \right) \right]}{2 \cos \left(\frac{x}{2} \right) \left[\cos \left(\frac{x}{2} \right) + \sin \left(\frac{x}{2} \right) \right]} = \frac{\sin \left(\frac{x}{2} \right)}{\cos \left(\frac{x}{2} \right)} = \tan \left(\frac{x}{2} \right)$$

The function elegantly collapses to:

$$f(x) = \ln(\sin(x)) \tan^2\left(\frac{x}{2}\right)$$

Differentiating using the product rule and chain rule:

$$f'(x) = \cot(x) \tan^2\left(\frac{x}{2}\right) + \ln(\sin(x)) \cdot 2 \tan\left(\frac{x}{2}\right) \cdot \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$f'(x) = \cot(x) \tan^2\left(\frac{x}{2}\right) + \ln(\sin(x)) \tan\left(\frac{x}{2}\right) \sec^2\left(\frac{x}{2}\right)$$

Evaluating at $x = \frac{\pi}{3}$. We know $\frac{x}{2} = \frac{\pi}{6}$. The required values are: $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$, $\cot\left(\frac{\pi}{3}\right) = \frac{1}{\sqrt{3}}$, $\tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}} \implies \tan^2\left(\frac{\pi}{6}\right) = \frac{1}{3}$, $\sec\left(\frac{\pi}{6}\right) = \frac{2}{\sqrt{3}} \implies \sec^2\left(\frac{\pi}{6}\right) = \frac{4}{3}$. Substitute these directly into the derivative:

$$f'\left(\frac{\pi}{3}\right) = \left(\frac{1}{\sqrt{3}}\right) \left(\frac{1}{3}\right) + \ln\left(\frac{\sqrt{3}}{2}\right) \left(\frac{1}{\sqrt{3}}\right) \left(\frac{4}{3}\right)$$

$$f'\left(\frac{\pi}{3}\right) = \frac{1}{3\sqrt{3}} + \frac{4}{3\sqrt{3}} \ln\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{9} \left(1 + 4 \ln\left(\frac{\sqrt{3}}{2}\right)\right)$$

Physics & Engineering

4.1 Mechanics

Spinning Cylinder in a Corner

A uniform cylinder of radius $R = \pi$ m is spun about its axis to an angular velocity $\omega_0 = 10 \text{ rad s}^{-1}$ and placed in a corner where the ground and wall are at 90° . The coefficient of kinetic friction is $\mu_k = 1$ between all surfaces. Find the number of turns the cylinder accomplishes before it stops. Take $g = 10 \text{ m s}^{-2}$.

Solution: Let the ground exert a normal force N_1 upwards and the wall exert a normal force N_2 horizontally. Because the cylinder is rotating against both surfaces, kinetic friction will oppose the motion at both contact points. $f_1 = \mu_k N_1 = N_1$, $f_2 = \mu_k N_2 = N_2$. From the balance of forces in the horizontal and vertical directions: $N_2 = f_1 = N_1$, $N_1 + f_2 = mg \implies N_1 + N_2 = mg$. Substituting $N_2 = N_1$ yields $2N_1 = mg$, meaning $N_1 = N_2 = \frac{mg}{2}$. The total frictional torque slowing down the cylinder is:

$$\tau = f_1 R + f_2 R = (N_1 + N_2) R = mgR$$

Using Newton's Second Law for rotation, $\tau = I\alpha$, where $I = \frac{1}{2}mR^2$ for a solid cylinder:

$$mgR = \left(\frac{1}{2}mR^2\right)\alpha \implies \alpha = \frac{2g}{R}$$

The time it takes to stop is $t = \frac{\omega_0}{\alpha} = \frac{\omega_0 R}{2g}$. The total angle rotated before stopping is:

$$\Delta\theta = \frac{\omega_0^2}{2\alpha} = \frac{\omega_0^2 R}{4g}$$

The number of turns is $N = \frac{\Delta\theta}{2\pi} = \frac{\omega_0^2 R}{8\pi g}$. Given $\omega_0 = 10$, $R = \pi$, and $g = 10$:

$$N = \frac{10^2 \pi}{8\pi(10)} = \frac{100}{80} = 1.25 \text{ turns}$$

4.2 Electric Circuits

RL Circuit Energy

A series RL circuit has inductance $L = 3 \text{ H}$ and resistance $R = 6 \Omega$. Immediately before $t = 0$ the current in the inductor is 5 A . At $t = 0$ a 12 V source is connected, its polarity driving current in the initial current reference direction. Find the energy stored in the inductor, in joules, at $t = \frac{\ln 3}{2} \text{ s}$.

Solution: The governing differential equation for a series RL circuit is $L \frac{di}{dt} + Ri = V$. Substituting the values:

$$3 \frac{di}{dt} + 6i = 12 \implies \frac{di}{dt} + 2i = 4$$

This is a standard first-order linear ordinary differential equation. The general solution is the sum of the steady-state and transient responses:

$$i(t) = i_{ss} + Ae^{-(\frac{R}{L})t} = \frac{12}{6} + Ae^{-2t} = 2 + Ae^{-2t}$$

We apply the initial condition $i(0) = 5$ A to determine A :

$$5 = 2 + A \implies A = 3$$

Thus, the current as a function of time is $i(t) = 2 + 3e^{-2t}$. We evaluate the current at $t = \frac{\ln 3}{2}$:

$$i\left(\frac{\ln 3}{2}\right) = 2 + 3e^{-2(\ln 3/2)} = 2 + 3e^{-\ln 3} = 2 + 3\left(\frac{1}{3}\right) = 2 + 1 = 3 \text{ A}$$

The energy stored in an inductor is $E = \frac{1}{2}Li^2$.

$$E = \frac{1}{2}(3)(3)^2 = \frac{27}{2} = 13.5 \text{ J}$$

4.3 Quantum Mechanics

Spin-1/2 Measurement Probability

Given the quantum state $|\psi\rangle_z = C\left(i\sqrt{3}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle\right)$, what is the probability that a measurement of the spin component m_y will yield $-1/2$? (Give your answer as a decimal, not as a percentage)

Solution: First, we find the normalization constant C :

$$C^2\left(|i\sqrt{3}|^2 + \left|-\frac{1}{\sqrt{2}}\right|^2\right) = 1 \implies C^2\left(3 + \frac{1}{2}\right) = 1 \implies C^2 = \frac{2}{7} \implies C = \sqrt{\frac{2}{7}}$$

The normalized state is $|\psi\rangle = \sqrt{\frac{6}{7}}i|\uparrow\rangle - \sqrt{\frac{1}{7}}|\downarrow\rangle$. We want the probability of measuring $m_y = -1/2$. The corresponding eigenstate is $|-y\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - i|\downarrow\rangle)$. The probability is given by $P = |\langle -y|\psi\rangle|^2$. We calculate the inner product:

$$\begin{aligned}\langle -y|\psi\rangle &= \frac{1}{\sqrt{2}}(\langle\uparrow| + i\langle\downarrow|)\left(\sqrt{\frac{6}{7}}i|\uparrow\rangle - \sqrt{\frac{1}{7}}|\downarrow\rangle\right) \\ &= \frac{1}{\sqrt{2}}\left(\sqrt{\frac{6}{7}}i - i\sqrt{\frac{1}{7}}\right) = \frac{i}{\sqrt{14}}(\sqrt{6} - 1)\end{aligned}$$

Squaring the magnitude gives the probability:

$$P = \left|\frac{i}{\sqrt{14}}(\sqrt{6} - 1)\right|^2 = \frac{1}{14}(\sqrt{6} - 1)^2 = \frac{1}{14}(6 - 2\sqrt{6} + 1) = \frac{7 - 2\sqrt{6}}{14}$$

Evaluating this to a decimal representation:

$$\frac{7 - 2(2.4495)}{14} = \frac{7 - 4.899}{14} = \frac{2.101}{14} = 0.15$$